UNIVERSITÀ DI PARMA · A.A. 2026/2027

Teoria e applicazioni delle reti complesse



Network Science

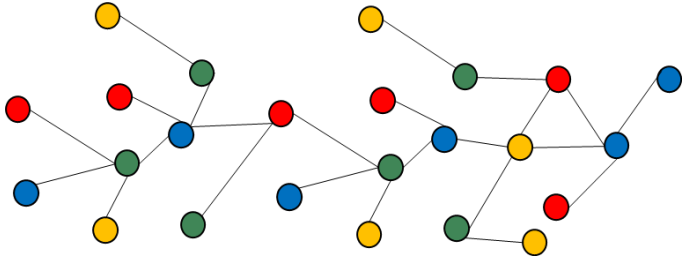
Michele Bellingeri

Laurea Magistrale in Scienze Informatiche · 1° anno · 6 CFU

Modulo 1

Introduzione

La **scienza delle reti** studia le **reti complesse**



Una **rete** è composta da:

Nodi ● (persone, luoghi, specie biologiche, computers…)

connessi da **legami (links)** ⟷

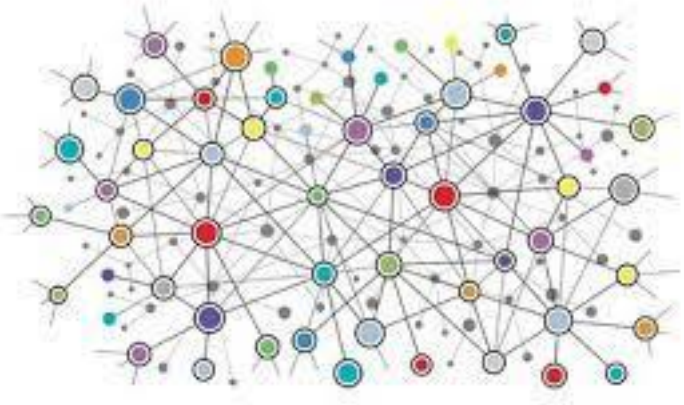
relazioni
contatti
flussi energia
strade,
connessioni
…

Dati Reali

Network : potente strumento per modellare sistemi complessi reali



pre-processing



Reti Reali

Reti Sociali

Reti che descrivono le interazioni sociali

Complex Networks

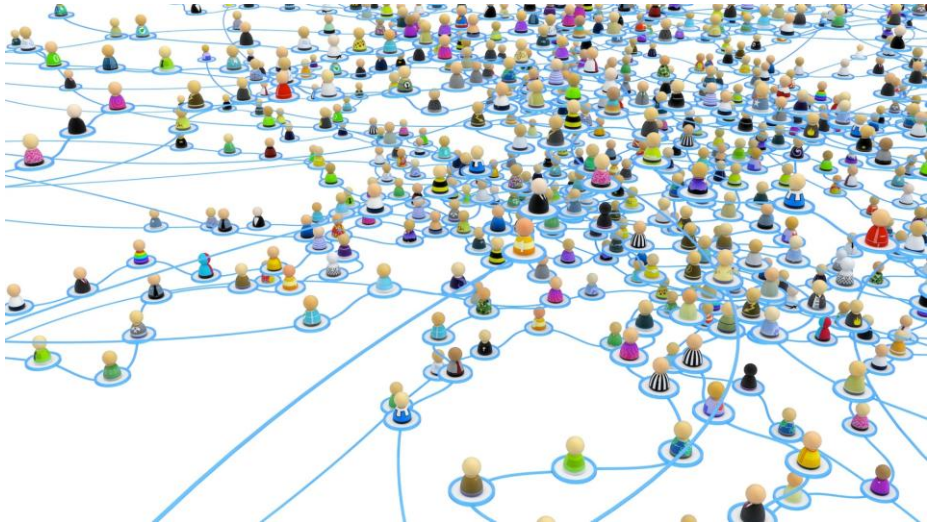
Nodi → Individui

Links → Amicizia

Contatto

Lavoro

Contagio



https://pmrpressrelease.com/global-location-based-social-networking-service-lbsns-market/

Reti Reali

Reti Sociali Online

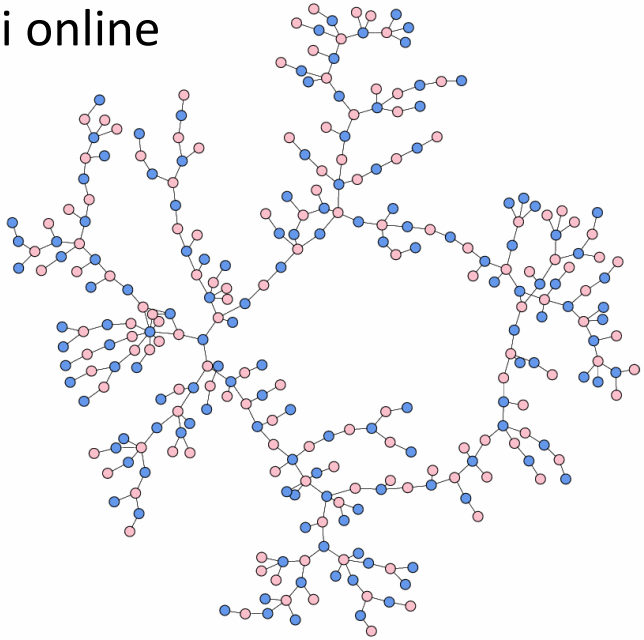
Social networks online descrivono le interazioni sociali online

Rete Complessa di Facebook

Nodi → Pagine

Links → Collegamento ipertestuale

→ ‘Mi piace’ reciproco

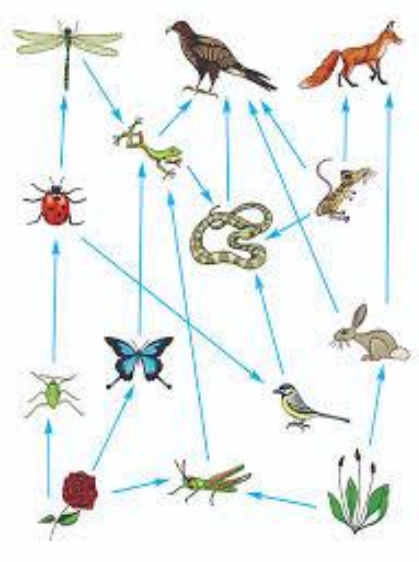


https://www.cs.uoi.gr/~tsap/teaching/2014-cs-l14/references.html

Reti Reali

Le reti ecologiche

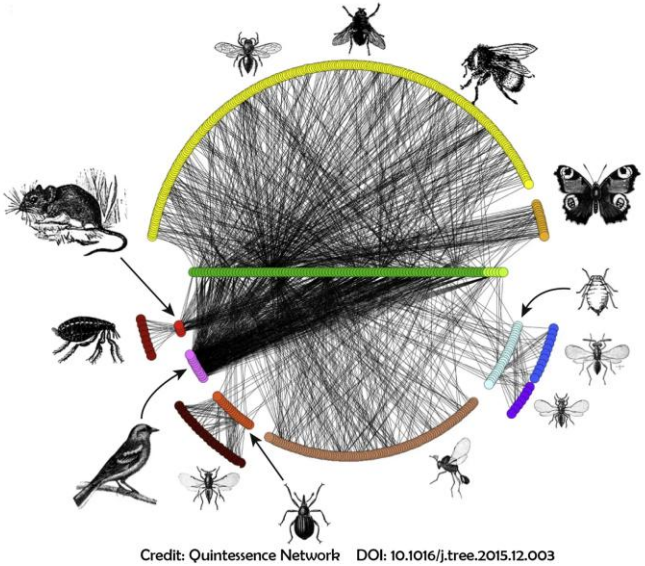
Reti trofiche descrivono '*chi mangia cosa*' negli ecosistemi



Complex Networks

Nodi → Specie

Link → Relazioni trofiche



Reti Reali

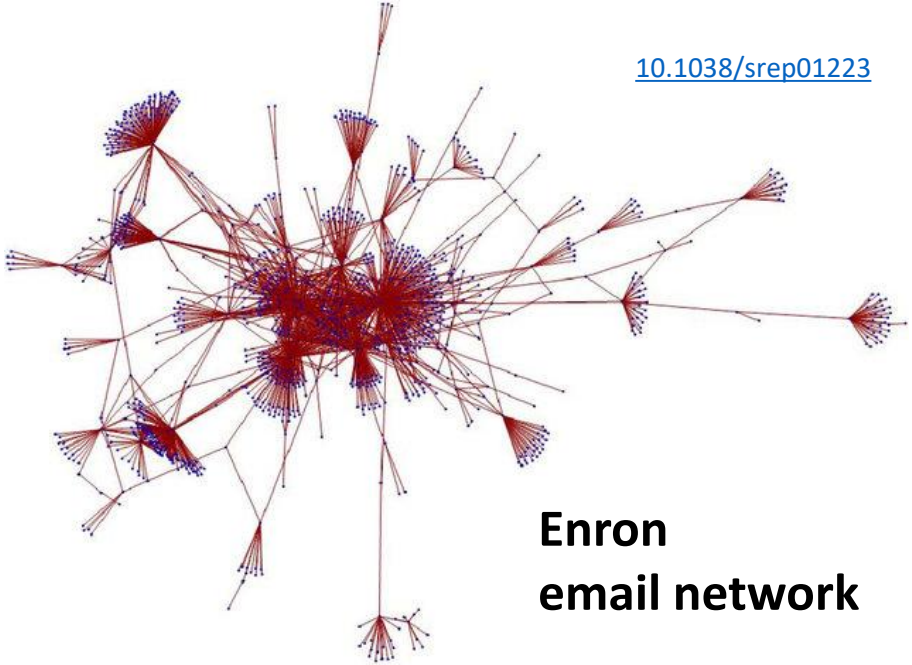
Reti di comunicazione

Modellano le attività di comunicazione tra individui o entità

Reti complesse di **e-mail**

Nodi ⟶ Impiegati

Link ⟶ Scambio di e-mail



10.1038/srep01223

Enron email network

Reti Reali

Reti di Computers

Nodi → Routers / PC

Link → Cavi



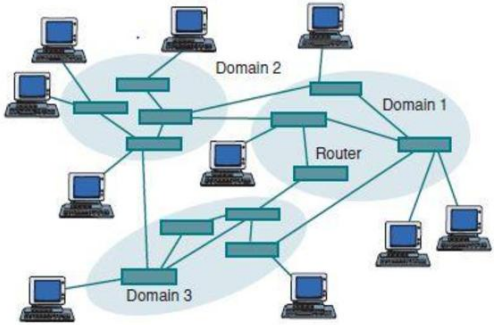
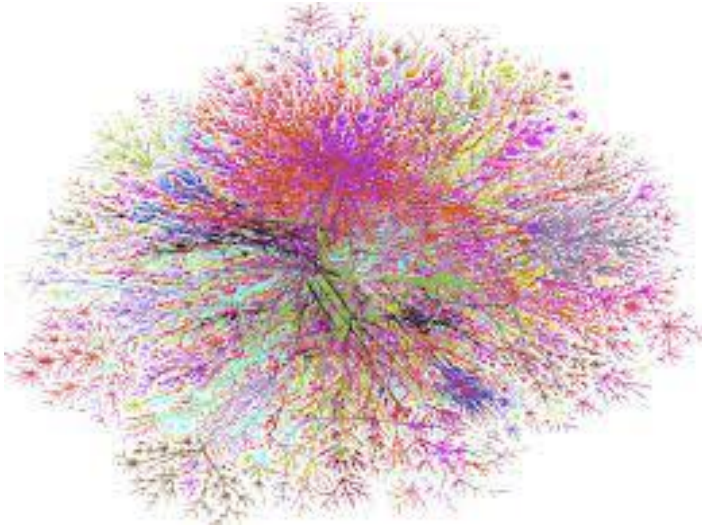


Fig. 1 – Network Structure of the Internet

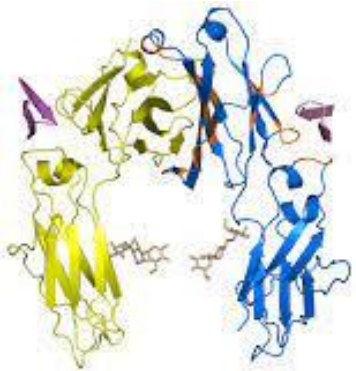
10.5120/17765-8882

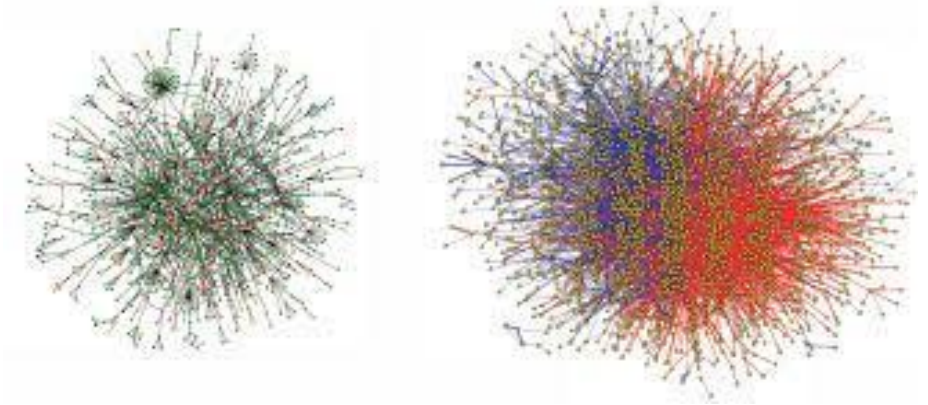


https://newmedialab.cuny.edu/project/complex-networks/

Reti Reali

Reti di interazione proteica

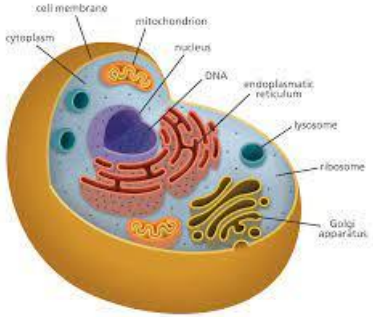




https://www.creative-proteomics.com/blog/index.php/brief-introduction-of-protein-protein-interaction-ppi/

Nodi ⟹ Proteine

Link ⟹ Interazioni Metaboliche



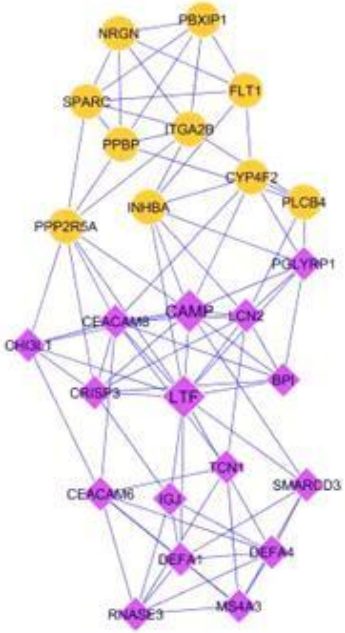
Reti Reali

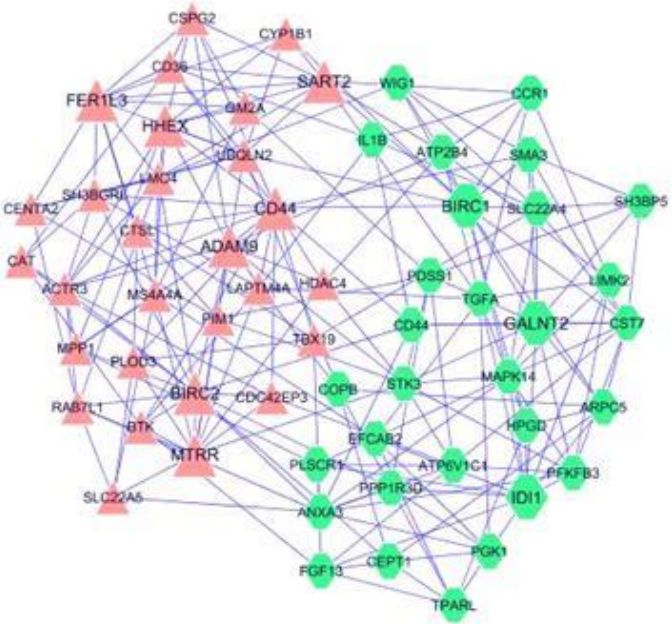


Rete di Regolazione Genica

Nodi ⟹ Geni
Fattori di Trascrizione

Link ⟹ Relazioni di regolazione





Un gene o una proteina regolatrice controlla l'espressione di un altro gene.

www.ese.wustl.edu/~nehorai/research/genomic/grn.html

Reti Reali



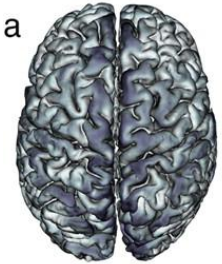
Rete cerebrale

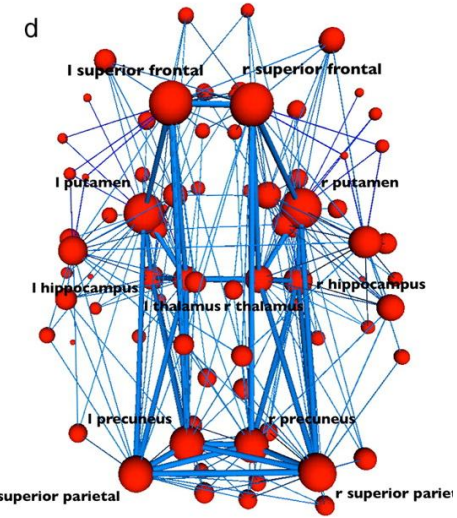
Nodi ⟹ Neuroni
Aree cerebrali

Link ⟹ Sinapsi
Fasci di fibre
Connessioni anatomiche
Attività Elettrica

~86 miliardi di neuroni

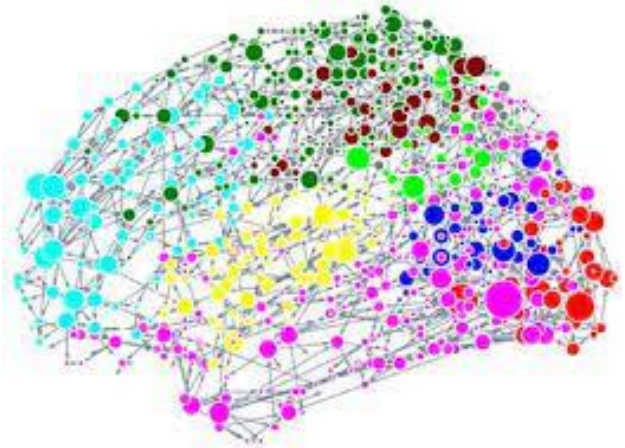
~10^{14}–10^{15} connessioni sinaptiche





10.1523/JNEUROSCI.3539-11.2011

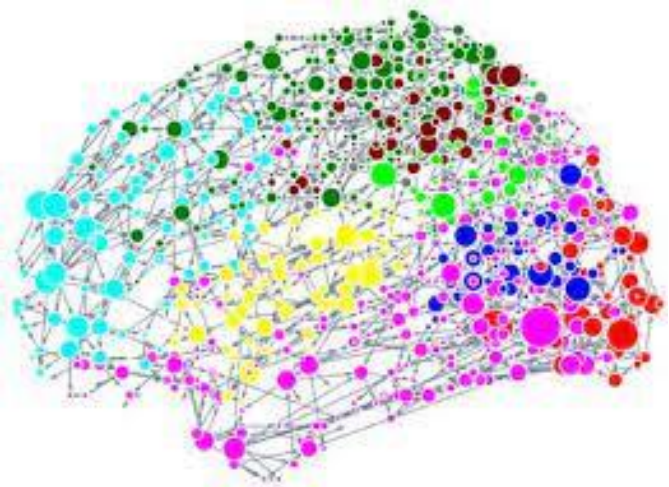
Human Connectome





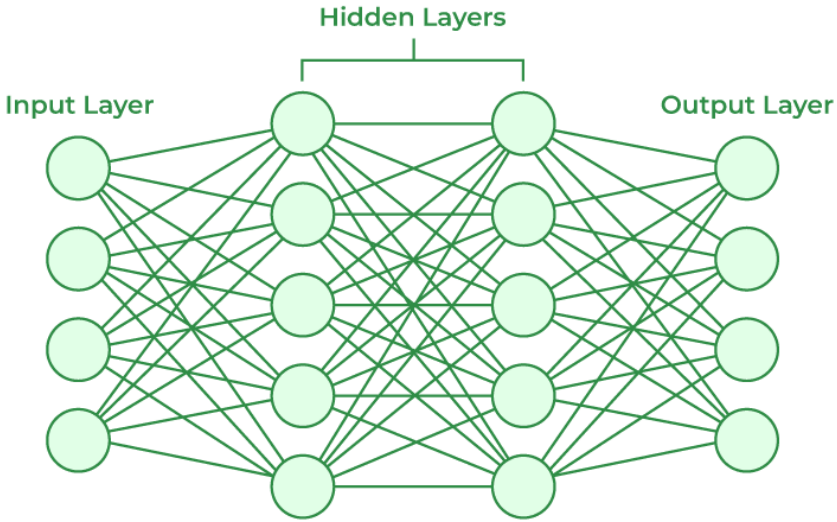
Reti Reali

Rete cerebrale



McCulloch e Pitts (1943) neurone come un'unità capace di ricevere diversi input e produrre un output

Reti Neurali



https://www.geeksforgeeks.org/deep-learning/artificial-neural-networks-and-its-applications/

Reti Reali

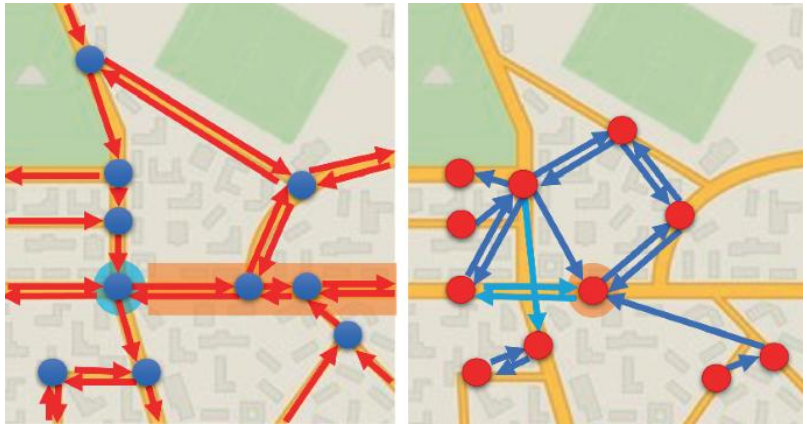
Rete stradale

Nodi ⟶ Incroci
Paesi
Città

Link ⟶ Strade







10.1007/978-3-031-15740-0_7

Reti Reali

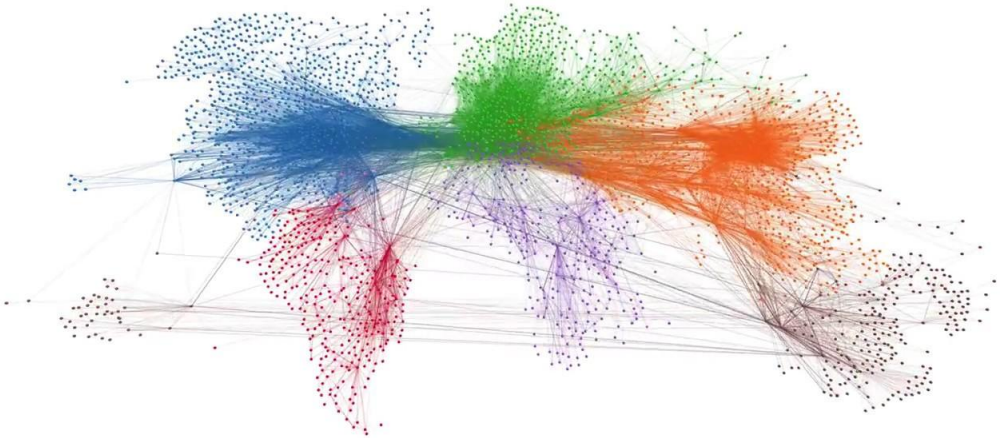
10.59521/5E2DDEC9FAD4593B



Rete aeroporti

Nodi ⟶ Aeroporti

Link ⟶ Voli
Rotte

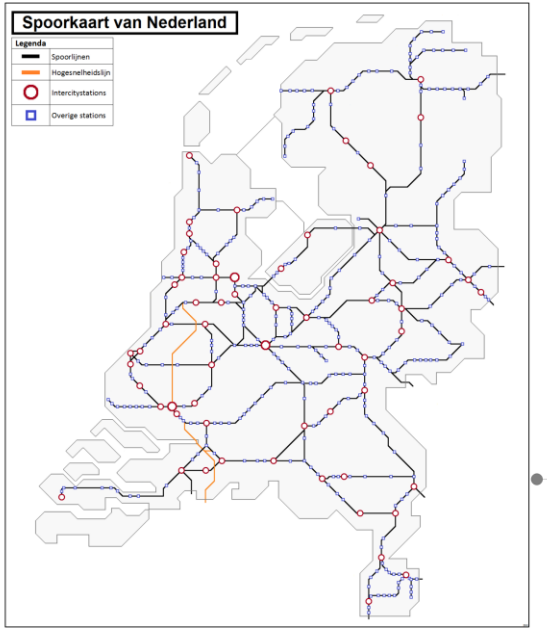


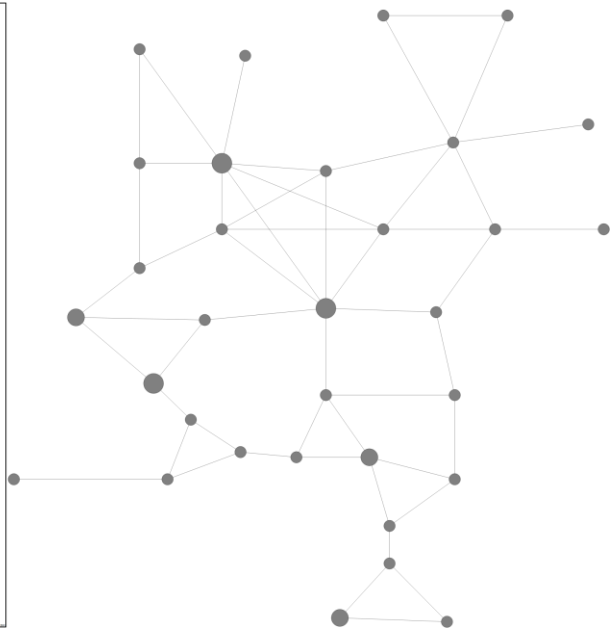
Reti Reali

Rete treni

Nodi ⟹ Stazioni

Link ⟹ Collegamenti





https://www.networkpages.nl/complex-networks/

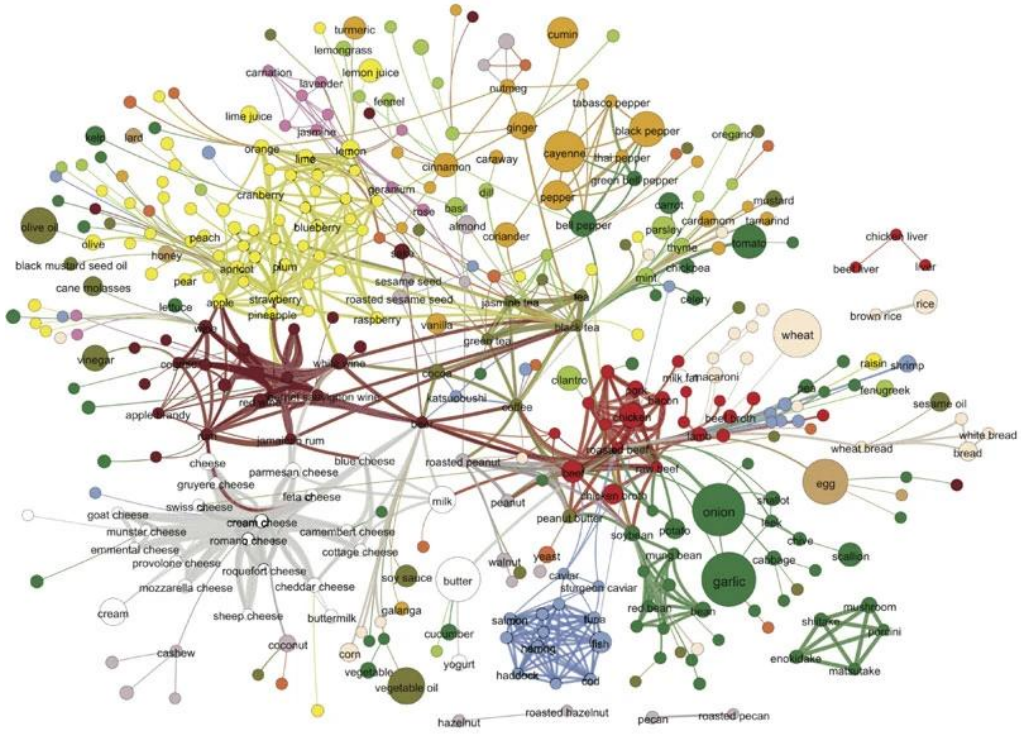
Reti Reali

10.1038/srep00196

Reti Gastronomiche Ingredienti

Nodi ⟶ Ingredienti

Link ⟶ Molecole comuni



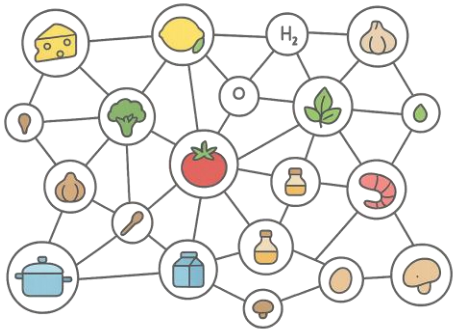
Reti Reali

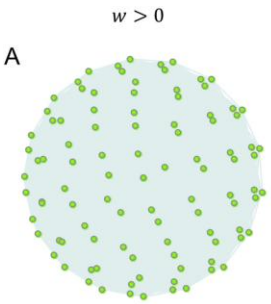
10.1038/s41598-025-17189-6

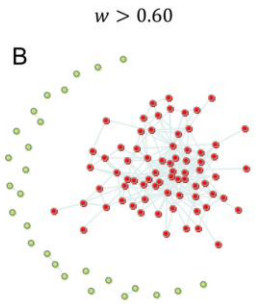
Reti Gastronomiche
Ricette

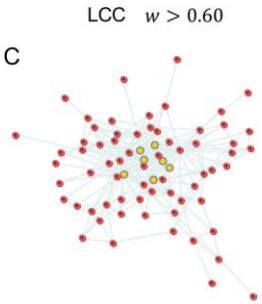
Nodi ⟶ Ricette

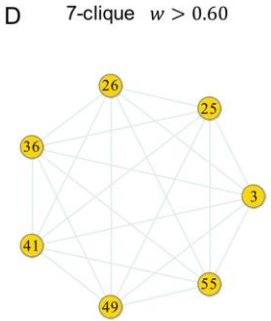
Link ⟶ Ingredienti comuni











E

Node 3 : *Allioli Sauce*

Node 25 : *Bean and Potato Stew*

Node 26 : *Mixed meat paella*

Node 36 : *Fairy rings mushrooms omelette*

Node 41 : *Monkfish Stew*

Node 49 : *Grilled Langoustines*

Node 55 : *Dry Salted Codfish Soup*

Reti Reali

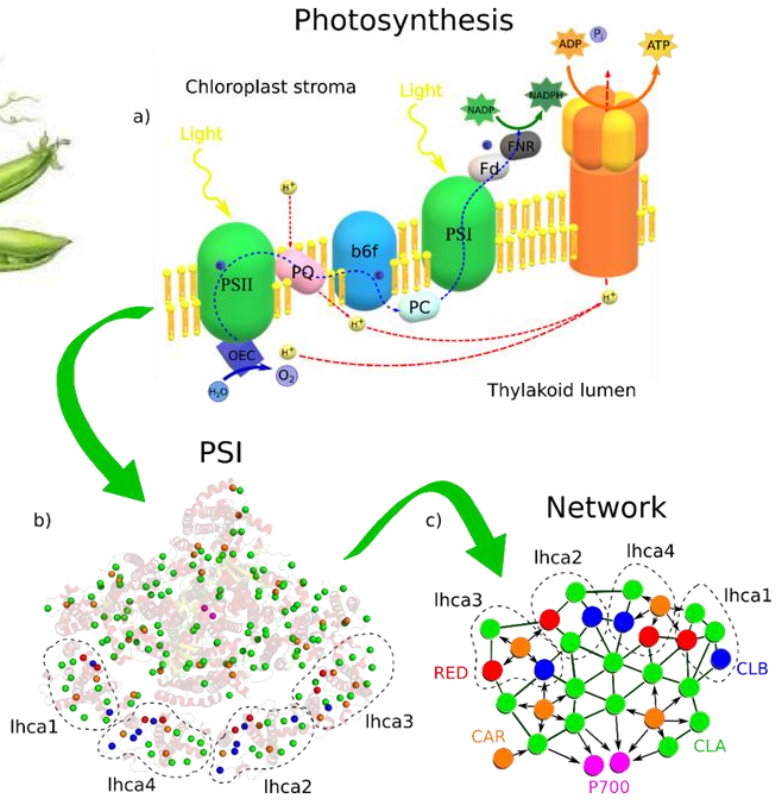
10.1098/rsif.2020.0813

Rete Fotosintetica

Nodi ⟹ Cromofori

Link ⟹ Trasferimenti energetici





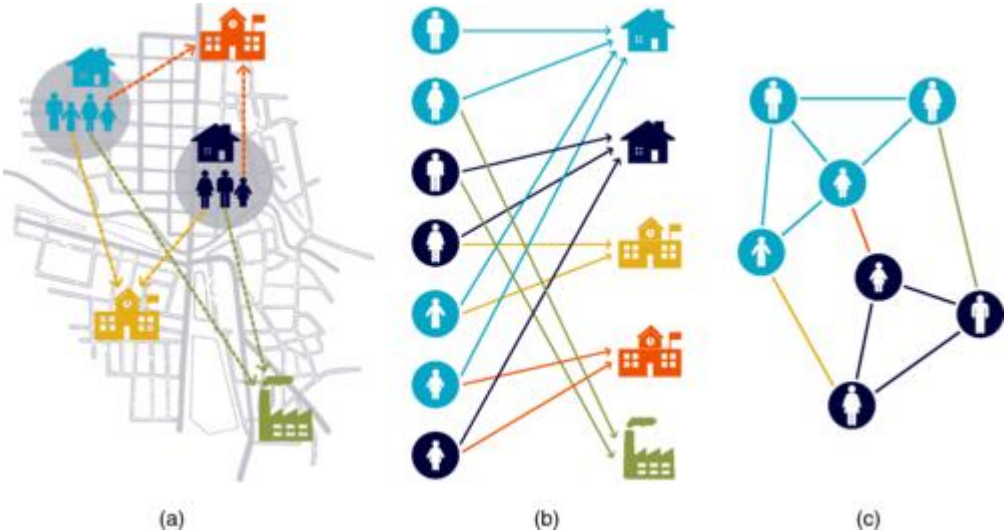
Reti Reali

Rete Epidemica

Nodi ⟶ Individui
Computers
Città
Animali
Aeroporti

Link ⟶ Contagio
interazioni sociali
contatto fisico
viaggi aerei
connessioni biologiche





Reti Reali

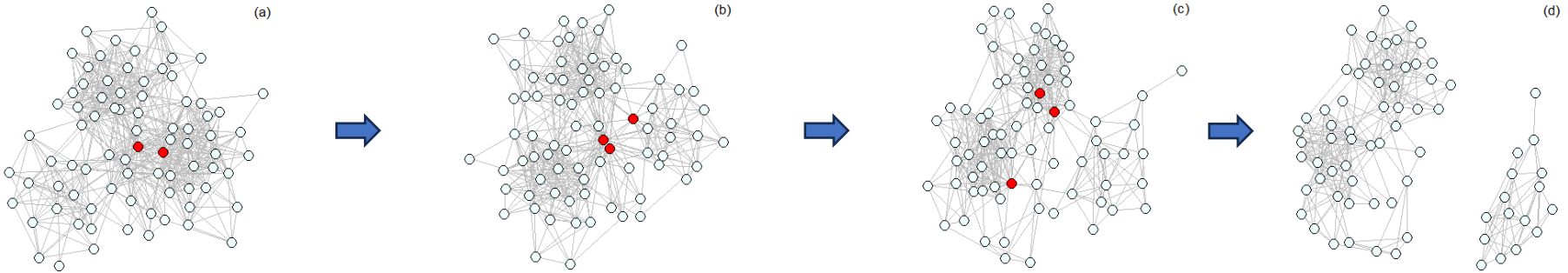


Rete contatti sociali

Rete interazioni animali

Rete di Computers

Rimozione di nodo ➡ **Vaccinazione di individuo**



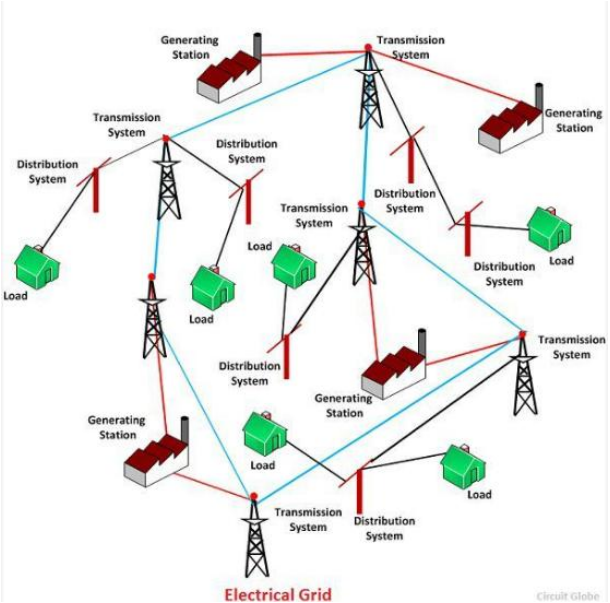
Vaccinazione di nodo ➡ Rompe canali di infezione ➡ Rallenta diffusione epidemica

Reti Reali

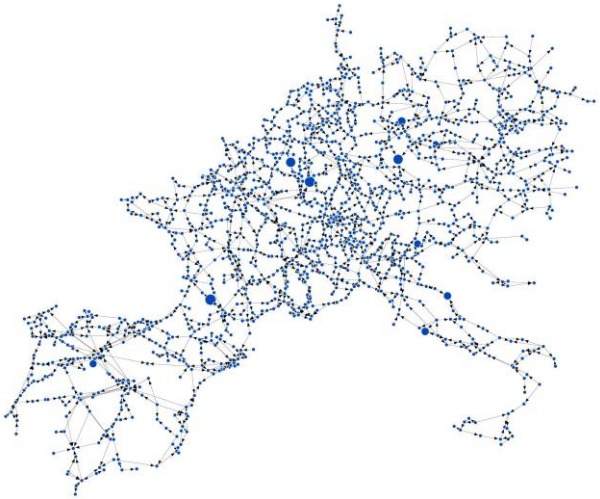
Rete Elettrica

Nodi → Centrali elettriche
Sottostazioni
Trasformatori
Città
Impianti rinnovabili

Link → Linee alta tensione
Cavi elettrici



https://www.electrical4u.com/electrical-grid-system/



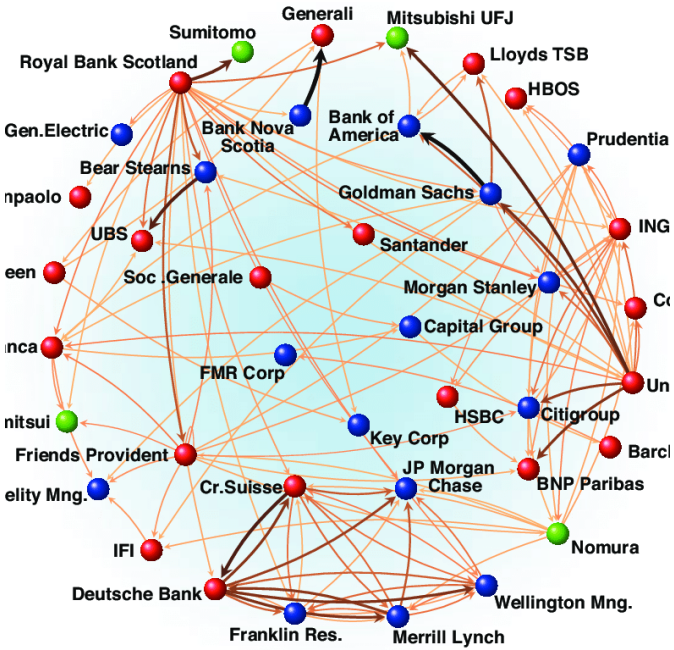
10.1038/srep04949

Reti Reali

Reti Finanziaria

Nodi ⟹ Banche
Stati
Aziende
Fondi di investimento
Mercati
Investitori

Link ⟹ Prestiti
Esposizioni finanziarie
Transazioni
Debiti
Partecipazioni azionarie



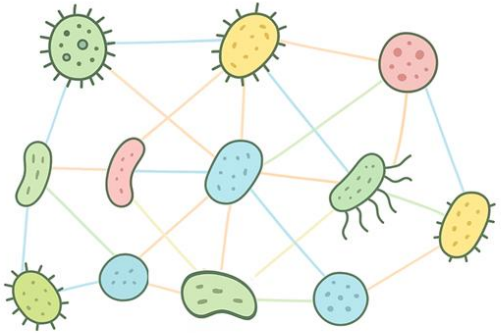
Rete finanziaria globale delle partecipazioni azionarie

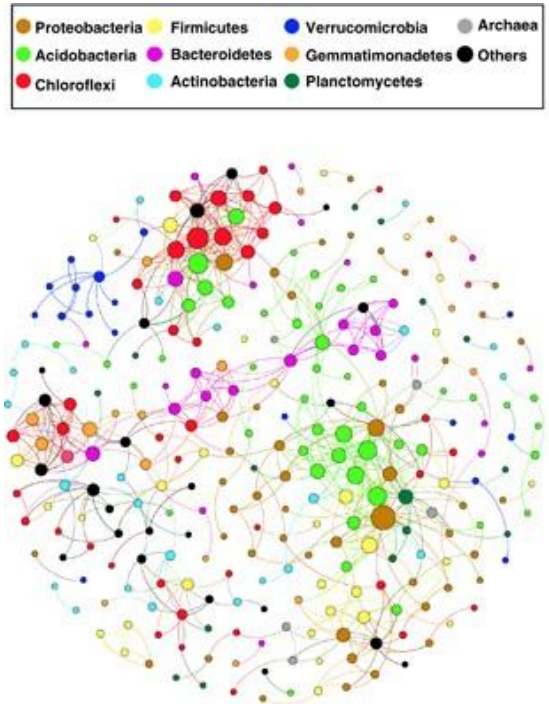
Reti Reali

Rete di Batteri

Nodi ⟹ Batteri

Link ⟹ Co-occorrenza
(campioni, ambienti, etc.)





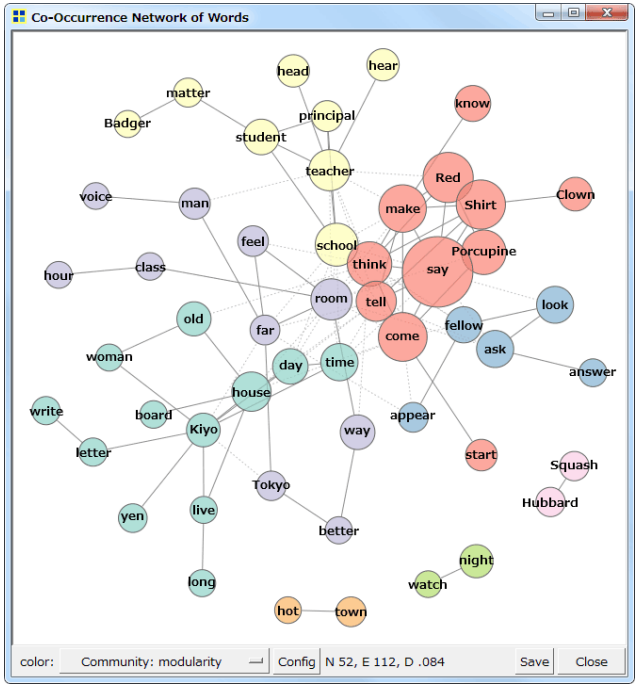
Reti Reali

Reti Linguistiche e Semantiche

Nodi ⟶ Parole

Link ⟶ Co-occorrenza
Relazioni grammaticali
Vicinanza nei testi
Similarità

Rete linguistica: nodi/parole e gli archi collegano parole che appaiono vicine nel testo.



https://en.wikipedia.org/wiki/Co-occurrence_network


Fine